Students

student_id (Primary Key), first_name ,last_name ,email ,enrollment year

Courses

course_id (Primary Key),course_name,department,credits

Instructors

instructor_id (Primary Key),first_name,last_name,email,department

Enrollments

enrollment_id (Primary Key), student_id (Foreign Key), course_id (Foreign Key), semester,year,grade

Course_Offerings

offering_id (Primary Key), course_id (Foreign Key), instructor_id (Foreign Key),Semester,Year

Table creating

CREATE TABLE Students (student_id INTEGER PRIMARY KEY, first_name TEXT, last_name TEXT, email TEXT, enrollment_year INTEGER);

CREATE TABLE Courses (course_id INTEGER PRIMARY KEY, course_name TEXT, department TEXT, credits INTEGER);

CREATE TABLE Instructors (instructor_id INTEGER PRIMARY KEY, first_name TEXT, last_name TEXT, email TEXT, department TEXT);

CREATE TABLE Enrollments (enrollment_id INTEGER PRIMARY KEY, student_id INTEGER, course_id INTEGER, semester TEXT, year INTEGER, grade TEXT, FOREIGN KEY(student_id) REFERENCES Students(student_id), FOREIGN KEY(course_id) REFERENCES Courses(course_id));

CREATE TABLE Course_Offerings (offering_id INTEGER PRIMARY KEY, course_id INTEGER, instructor_id INTEGER, semester TEXT, year INTEGER, FOREIGN KEY(course_id) REFERENCES Courses(course_id), FOREIGN KEY(instructor_id) REFERENCES Instructors(instructor_id));

Data Input
INSERT INTO Students VALUES

(1, 'John', 'Doe', 'john.doe@email.com', 2022),

(2, 'Jane', 'Smith', 'jane.smith@email.com', 2023),

(3, 'Alice', 'Johnson', 'alice.j@email.com', 2024),

(4, 'Bob', 'Williams', 'bob.w@email.com', 2023),

(5, 'Charlie', 'Brown', 'charlie.b@email.com', 2022);

INSERT INTO Courses VALUES

(1, 'Intro to Programming', 'Computer Science', 3),

(2, 'Data Structures', 'Computer Science', 4),

(3, 'Calculus I', 'Mathematics', 4),

(4, 'English Literature', 'Humanities', 3),

(5, 'Databases', 'Computer Science', 3);

INSERT INTO Instructors VALUES

(1, 'Dr. Alan', 'Turing', 'alan.t@email.com', 'Computer Science'),

(2, 'Dr. Marie', 'Curie', 'marie.c@email.com', 'Mathematics'),

(3, 'Dr. Jane', 'Austen', 'jane.a@email.com', 'Humanities'),

(4, 'Dr. Ada', 'Lovelace', 'ada.l@email.com', 'Computer Science'),

(5, 'Dr. Isaac', 'Newton', 'isaac.n@email.com', 'Mathematics');

INSERT INTO Course_Offerings VALUES

(1, 1, 1, 'Fall', 2024),

(2, 2, 4, 'Spring', 2024),

(3, 3, 2, 'Fall', 2024),

(4, 4, 3, 'Spring', 2024),

(5, 5, 1, 'Spring', 2024),

(6, 2, 1, 'Fall', 2023),

(7, 5, 4, 'Fall', 2024);

INSERT INTO Enrollments VALUES

(1, 1, 1, 'Fall', 2024, 'A'),

(2, 2, 2, 'Spring', 2024, 'B'),

(3, 3, 3, 'Fall', 2024, 'C'),

(4, 4, 1, 'Fall', 2024, 'A'),

(5, 5, 5, 'Spring', 2024, 'B'),

(6, 1, 2, 'Fall', 2023, 'A'),

(7, 2, 5, 'Fall', 2024, 'C')

(8, 3, 2, 'Fall', 2023, 'B');

Q 1: List all students enrolled in the year 2023.
Solve: SELECT * FROM Students WHERE enrollment_year = 2023;

Q 2: Retrieve all courses offered by the 'Computer Science' department.

Solve: SELECT * FROM Courses WHERE department = 'Computer Science';

Q 3: Find the total number of students enrolled in each course in Fall 2024.

Solve: SELECT c.course_name, COUNT(e.student_id) AS student_count FROM Enrollments e JOIN Courses c ON e.course_id = c.course_id WHERE e.semester = 'Fall' AND e.year = 2024 GROUP BY c.course_id;

Q 4: Get the names of instructors who have taught more than one course.

Solve: SELECT i.first_name, i.last_name FROM Course_Offerings co JOIN Instructors i ON co.instructor_id = i.instructor_id GROUP BY co.instructor_id HAVING COUNT(DISTINCT co.course_id) > 1;

Q 5: List all students who scored an 'A' in any course.

Solve: SELECT DISTINCT s.first_name, s.last_name FROM Enrollments e JOIN Students s ON e.student_id = s.student_id WHERE e.grade = 'A';

Q 6: Get the average grade (numerical value, assume A=4, B=3, etc.) for each student.

Solve: SELECT s.first_name, s.last_name, ROUND(AVG(CASE grade WHEN 'A' THEN 4 WHEN 'B' THEN 3 WHEN 'C' THEN 2 END), 2) AS avg_grade FROM Enrollments e JOIN Students s ON e.student_id = s.student_id GROUP BY e.student_id;

Q 7: List courses that have never been offered.

Solve: SELECT c.course_name FROM Courses c LEFT JOIN Course_Offerings co ON c.course_id = co.course_id WHERE co.course_id IS NULL;

Q 8: Find the names of students who have not enrolled in any course.

Solve: SELECT s.first_name, s.last_name FROM Students s LEFT JOIN Enrollments e ON s.student_id = e.student_id WHERE e.student_id IS NULL;

Q 9: Retrieve a list of courses and the names of instructors teaching them in Spring 2024.

Solve: SELECT c.course_name, i.first_name || ' ' || i.last_name AS instructor FROM Course_Offerings co JOIN Courses c ON co.course_id = c.course_id JOIN Instructors i ON co.instructor_id = i.instructor_id WHERE co.semester = 'Spring' AND co.year = 2024;

Q 10: List all courses along with the number of students enrolled in each.
Solve: SELECT c.course_name, COUNT(e.student_id) AS total_enrolled FROM Courses c LEFT JOIN Enrollments e ON c.course_id = e.course_id GROUP BY c.course_id;